

Nonlinear Equations

Nonlinear Equations

- ▶ Given function f , we seek value x for which

$$f(x) = 0$$

- ▶ Solution x is *root* of equation, or *zero* of function f
- ▶ So problem is known as *root finding* or *zero finding*

Existence, Uniqueness, and Conditioning

Existence and Uniqueness

- ▶ Existence and uniqueness of solutions are more complicated for nonlinear equations than for linear equations
- ▶ For function $f: \mathbb{R} \rightarrow \mathbb{R}$, *bracket* is interval $[a, b]$ for which sign of f differs at endpoints
- ▶ If f is continuous and $\text{sign}(f(a)) \neq \text{sign}(f(b))$, then Intermediate Value Theorem implies there is $x^* \in [a, b]$ such that $f(x^*) = 0$
- ▶ There is no simple analog for n dimensions

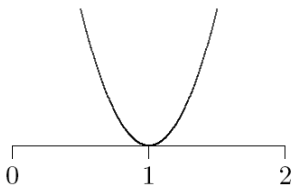
Examples: One Dimension

Nonlinear equations can have any number of solutions

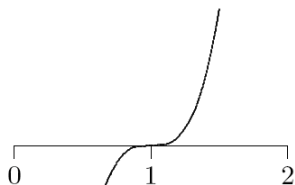
- ▶ $\exp(x) + 1 = 0$ has no solution
- ▶ $\exp(-x) - x = 0$ has one solution
- ▶ $x^2 - 4\sin(x) = 0$ has two solutions
- ▶ $x^3 + 6x^2 + 11x - 6 = 0$ has three solutions
- ▶ $\sin(x) = 0$ has infinitely many solutions

Multiplicity

- If $f(x^*) = f'(x^*) = f''(x^*) = \dots = f^{(m-1)}(x^*) = 0$ but $f^{(m)}(x^*) \neq 0$ (i.e., m th derivative is lowest derivative of f that does not vanish at x^*), then root x^* has **multiplicity** m



$$x^2 - 2x + 1$$



$$x^3 - 3x^2 + 3x - 1$$

- If $m = 1$ ($f(x^*) = 0$ and $f'(x^*) \neq 0$), then x^* is **simple** root

Convergence of Iterative Methods

Convergence Rate

- ▶ For general iterative methods, define error at iteration k by

$$\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$$

where $\mathbf{x}_k$ is approximate solution and $\mathbf{x}^*$ is true solution

- ▶ For methods that maintain interval known to contain solution, rather than specific approximate value for solution, take error to be length of interval containing solution
- ▶ Sequence converges with rate r if

$$\lim_{k \rightarrow \infty} \frac{\|\mathbf{e}_{k+1}\|}{\|\mathbf{e}_k\|^r} = C$$

for some finite nonzero constant C

Convergence Rate, continued

Some particular cases of interest

- ▶ $r = 1$: *linear* ($C < 1$)
- ▶ $r > 1$: *superlinear*
- ▶ $r = 2$: *quadratic*

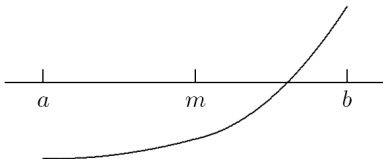
Convergence rate	Digits gained per iteration
linear	constant
superlinear	increasing
quadratic	double

Bisection Method in 1D

Interval Bisection Method

Bisection method begins with initial bracket and repeatedly halves its length until solution has been isolated as accurately as desired

```
while  $((b - a) > tol)$  do  
     $m = a + (b - a)/2$   
    if  $\text{sign}(f(a)) = \text{sign}(f(m))$  then  
         $a = m$   
    else  
         $b = m$   
    end  
end
```



Example: Bisection Method

$$f(x) = x^2 - 4 \sin(x) = 0$$

a	$f(a)$	b	$f(b)$
1.000000	-2.365884	3.000000	8.435520
1.000000	-2.365884	2.000000	0.362810
1.500000	-1.739980	2.000000	0.362810
1.750000	-0.873444	2.000000	0.362810
1.875000	-0.300718	2.000000	0.362810
1.875000	-0.300718	1.937500	0.019849
1.906250	-0.143255	1.937500	0.019849
1.921875	-0.062406	1.937500	0.019849
1.929688	-0.021454	1.937500	0.019849
1.933594	-0.000846	1.937500	0.019849
1.933594	-0.000846	1.935547	0.009491
1.933594	-0.000846	1.934570	0.004320
1.933594	-0.000846	1.934082	0.001736
1.933594	-0.000846	1.933838	0.000445

Bisection Method, continued

- ▶ Bisection method makes no use of magnitudes of function values, only their signs
- ▶ Bisection is certain to converge, but does so slowly
- ▶ At each iteration, length of interval containing solution reduced by half, convergence rate is *linear*, with $r = 1$ and $C = 0.5$
- ▶ One bit of accuracy is gained in approximate solution for each iteration of bisection
- ▶ Given starting interval $[a, b]$, length of interval after k iterations is $(b - a)/2^k$, so achieving error tolerance of tol requires

$$\left\lceil \log_2 \left(\frac{b - a}{tol} \right) \right\rceil$$

iterations, regardless of function f involved

Fixed-Point Iteration in 1D

Fixed-Point Problems

- ▶ *Fixed point* of given function $g: \mathbb{R} \rightarrow \mathbb{R}$ is value x such that

$$x = g(x)$$

- ▶ Many iterative methods for solving nonlinear equations use *fixed-point iteration* scheme of form

$$x_{k+1} = g(x_k)$$

where fixed points for g are solutions for $f(x) = 0$

- ▶ Also called *functional iteration*, since function g is applied repeatedly to initial starting value x_0
- ▶ For given equation $f(x) = 0$, there may be many equivalent fixed-point problems $x = g(x)$ with different choices for g

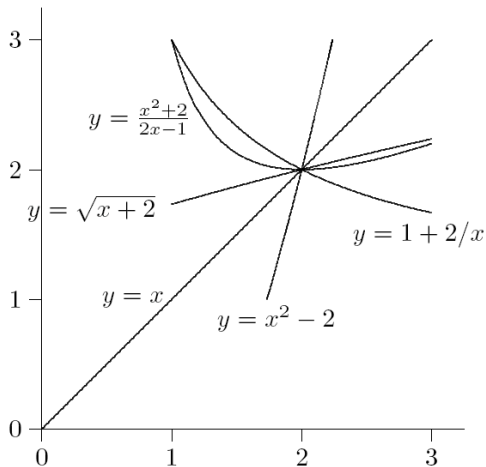
Example: Fixed-Point Problems

If $f(x) = x^2 - x - 2$, then fixed points of each of functions

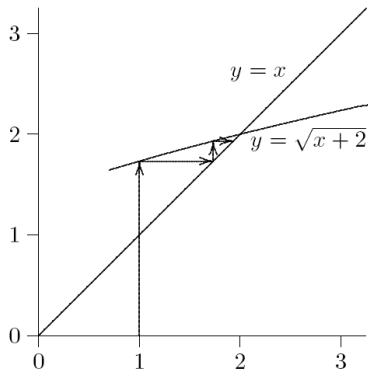
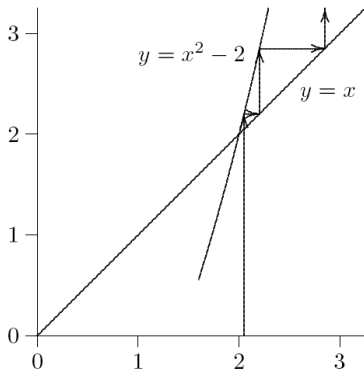
- ▶ $g(x) = x^2 - 2$
- ▶ $g(x) = \sqrt{x + 2}$
- ▶ $g(x) = 1 + 2/x$
- ▶ $g(x) = \frac{x^2 + 2}{2x - 1}$

are solutions to equation $f(x) = 0$

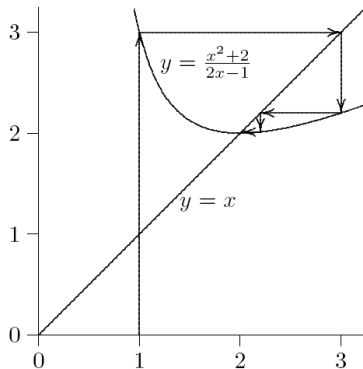
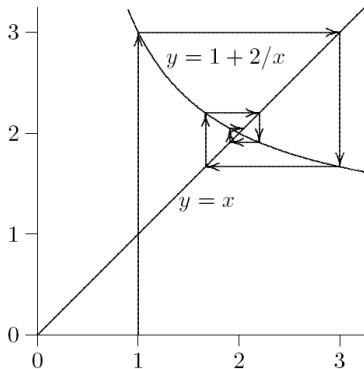
Example: Fixed-Point Problems



Example: Fixed-Point Iteration



Example: Fixed-Point Iteration



Convergence of Fixed-Point Iteration

- ▶ If $x^* = g(x^*)$ and $|g'(x^*)| < 1$, then there is interval containing x^* such that iteration

$$x_{k+1} = g(x_k)$$

converges to x^* if started within that interval

- ▶ If $|g'(x^*)| > 1$, then iterative scheme diverges
- ▶ Asymptotic convergence rate of fixed-point iteration is usually linear, with constant $C = |g'(x^*)|$
- ▶ But if $g'(x^*) = 0$, then convergence rate is at least quadratic

Newton's Method in 1D

Newton's Method

- ▶ Truncated Taylor series

$$f(x + h) \approx f(x) + f'(x)h$$

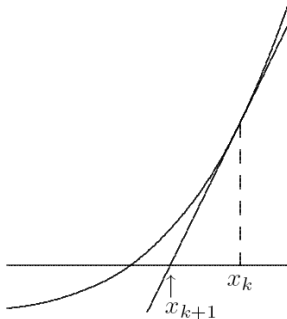
is linear function of h approximating f near x

- ▶ Replace nonlinear function f by this linear function, whose zero is $h = -f(x)/f'(x)$
- ▶ Zeros of original function and linear approximation are not identical, so repeat process, giving *Newton's method*

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Newton's Method, continued

Newton's method approximates nonlinear function f near x_k by *tangent line* at $f(x_k)$



Example: Newton's Method

- Use Newton's method to find root of

$$f(x) = x^2 - 4 \sin(x) = 0$$

- Derivative is

$$f'(x) = 2x - 4 \cos(x)$$

so iteration scheme is

$$x_{k+1} = x_k - \frac{x_k^2 - 4 \sin(x_k)}{2x_k - 4 \cos(x_k)}$$

- Taking $x_0 = 3$ as starting value, we obtain

x	$f(x)$	$f'(x)$	h
3.000000	8.435520	9.959970	-0.846942
2.153058	1.294772	6.505771	-0.199019
1.954039	0.108438	5.403795	-0.020067
1.933972	0.001152	5.288919	-0.000218
1.933754	0.000000	5.287670	0.000000

Convergence of Newton's Method

- ▶ Newton's method transforms nonlinear equation $f(x) = 0$ into fixed-point problem $x = g(x)$, where

$$g(x) = x - f(x)/f'(x)$$

and hence

$$g'(x) = f(x)f''(x)/(f'(x))^2$$

- ▶ If x^* is simple root (i.e., $f(x^*) = 0$ and $f'(x^*) \neq 0$), then $g'(x^*) = 0$
- ▶ Convergence rate of Newton's method for simple root is therefore *quadratic* ($r = 2$)
- ▶ But iterations must start close enough to root to converge

Newton's Method, continued

For multiple root, convergence rate of Newton's method is only linear, with constant $C = 1 - (1/m)$, where m is multiplicity

k	$f(x) = x^2 - 1$	$f(x) = x^2 - 2x + 1$
0	2.0	2.0
1	1.25	1.5
2	1.025	1.25
3	1.0003	1.125
4	1.00000005	1.0625
5	1.0	1.03125